

Quiz 4 - Part I

BLG 483E - Artificial Intelligence Aided Computer Engineering
CRN: 25311 - 25410

Data Consent: I agree that my anonymized responses in this assignment may be used for academic research and educational purposes.

Name / ID: _____

Date: _____

1 Part 1: Academic Background and Course Preferences

To map the relationship between subject difficulty and effective communication strategies, please fill in the tables below.

1.1 Perceived Strengths (Top 3 Courses I Found Easier to Understand)

This section reflects courses that I found relatively easier to understand compared to my peers, along with my performance.

Course Name	Grade	Reason (Why it felt easier)
1.		
2.		
3.		

1.2 Perceived Challenges (Top 3 Courses I Found Difficult to Understand)

This section focuses on courses that I found conceptually challenging, regardless of my final grade.

Course Name	Grade	Reason (What made it difficult)
1.		
2.		
3.		

Note: The evaluation considers both academic performance and perceived conceptual difficulty.

2 Part 2: Content Generation and Ninova Repository

Choose a specific sub-topic from your "Difficult Courses" list. Your task is to use AI to generate a slide deck that explains this topic in a way that you find most effective for learning.

Mandatory Task: The Generative Loop

1. **Topic Selection:** Pick a concept you found hard to grasp (e.g., Eigenvalues, Paging, Red-Black Trees) and briefly explain *why* this topic was difficult for you.
2. **Generation:** Ask the AI to produce 6-8 LaTeX Beamer slides using **any method or style** you think best explains the topic.
3. **Iterative Refinement:** Review the generated slides. If the explanation is unclear, incomplete, or incorrect, refine your prompt and regenerate the slides.
 - **Iteration Rule:** You are required to perform at least 3 and at most 5 iterations. (e.g., "Explain the core concept of this slide as if you are talking to a 10-year-old" to simplify.)
 - **Version Tracking:** You must upload each intermediate version (e.g., v2.tex, v3.tex) to Ninova as you progress.
 - **Goal:** Each refinement should aim for a higher pedagogical quality (e.g., adding a TikZ diagram, simplifying a proof, or adding real-world examples).
4. **NINOVA UPLOAD:** Compile your *final revised* slides into a LaTeX and upload the file to the Ninova.

2.1 Prompt Engineering Log

- **Initial Prompt:** Paste your first command to the AI here.
- **Design Choice:** Why did you choose this specific way of explaining the topic?
- **Iterative Refinement Process:** Describe how you improved the AI output. If the initial result was unclear, incomplete, or incorrect, explain how you refined your prompt. Indicate how many iterations you performed and what changes you made at each step.
- **Before vs After:** Briefly compare the initial AI output with the final version. What were the main weaknesses of the first output, and how were they improved in the final result?

3 Part 3: Comparative Performance Analysis

Compare the instructor's provided samples (*Chinese Room*, *Rice's Theorem*, *Convolution*) with your own generated output.

1. **Communication Strategy:** Which instructor slide was most effective for you? Identify the key elements (e.g., step-by-step explanation, visuals, examples) and explain why they worked. Briefly compare with your own slide(s).
2. **Information Density:** Choose one of your slides. If you added a TikZ diagram, what would your prompt be? Specify which part is unclear without it and how the diagram improves understanding.
3. **Efficiency:** Which part of your slides does the AI explain better than a textbook? Explain why (e.g., clearer examples, better abstraction, step-by-step structure) and briefly compare.
4. **Slide Improvement Challenge:** Select the most effective instructor slide and write a refined prompt to improve it (structure, explanation style, visuals). Mention the prompting technique(s) used.
5. **Missing Piece:** Identify a key missing element (e.g., intuition, real-world context, edge cases) in either set of slides and explain its impact on understanding.

4 Part 4: Quality Assurance and Interactive Testing

As a "Content Architect," you must ensure your Ninova submission is actually "learnable."

- a) **The Verification Quiz:** Write two questions based **only** on the slides you generated.
- Question 1: Must require the reader to interpret a graphic, table, or code snippet from your slides.
 - Question 2: Must require a logical derivation or calculation based on your slides, and should be non-trivial.
 - For each question, briefly state the expected answer or explain why the answer is correct.
- b) **The Decompression Task:** Identify the most complex slide you generated. Write a prompt that forces the AI to "split" that slide into a 3-step sequence to make it easier for a beginner.

5 Part 5: Feedback on AI Synthesis

- Which elements were the most difficult to get the AI to generate correctly? Explain *why* these elements were difficult (e.g., ambiguity, lack of clarity, or visualization challenges).
- What types of mistakes or weaknesses did the AI repeatedly show during the generation process (e.g., overly verbose explanations, missing examples, incorrect abstraction)?
- If you were to give a single "Golden Rule" prompt to a friend who is using AI to study for a difficult exam, what would it be?